Final Activity. DC-DC Power Converters

Objective: To analyze and understand the operating principles, mathematical models, and practical applications of three fundamental DC-DC converter topologies Buck, Boost, and Buck-Boost through a technical presentation, a brief literature review, and a computational simulation using MATLAB or Simulink.

Instructions

1. Prepare a 10-slide presentation entirely in English explaining the three DC-DC converter topologies: Buck, Boost, and Buck-Boost. The presentation must include:
    a. The operating principles and main characteristics of each converter.
    b. Electrical circuit diagrams identifying the main components.
    c. Mathematical equations describing the operation and voltage conversion relationships of all three topologies.

    The slides must be visually innovative, and technically accurate. You could use circuit diagrams, illustrations, graphs, animations, and other visual resources to support your explanations. Slides should maintain a proper balance between technical content and visual elements, avoiding excessive text.

2. Research for three scientific articles, one related to each converter topology. The selected articles should describe a specific engineering problem or practical application involving the corresponding converter. You must prepare a brief literature review in PDF format, including the following information for each article.
    a. Article information: Title, authors, publication year, and journal.
    b. Engineering problem**:** What specific problem does the research address?
    c. Converter application: How is the converter topology used to solve the problem?
    d. Main findings: What are the most relevant results or conclusions of the research?

    Recommended length: Approximately one page per article.

3. You must select only one of the three converter topologies, Buck, Boost, or Buck-Boost and develop a basic computational simulation using MATLAB or Simulink. The simulation must demonstrate the converter's operating behavior and generate graphical results.
    a. Define the main simulation parameters, including input voltage, duty cycle, and relevant circuit components.
    b. Implement a mathematical or circuit-based model of the selected converter.
    c. Generate at least two relevant plots, including the output voltage and another electrical variable, such as inductor current.
    d. Analyze the converter's behavior based on the simulation results.

    You must submit a brief technical report in PDF format explaining: the selected converter and the objective of the simulation, the mathematical model or simulation diagram, the parameters and conditions used in the simulation and the graphical results obtained. The report must demonstrate that you understand the simulated behavior.

Deliverables

- 10-slides presentation in pdf.
- 3 research papers consulted that address Buck, Boost and Buck-Boost topologies.
- A summary in pdf of the above research papers.
- Matlab script (.m) or Simulink model (.mdl) of the simulation.
- Simulation report, including graphs and explanations.